Matrices: Formula Sheet

1. Basics

Order: $m \times n$, elements mn

Trace: $\operatorname{tr} A = \sum a_{ii}$

Scalar matrix: $|kI_n| = k^n$

Diagonal: $D^m = \operatorname{diag}(d_i^m)$

Symmetric matrices: $s^{n(n+1)/2}$

2. Operations

Sum: $A \pm B = [a_{ij} \pm b_{ij}]$

Trace:

$$\operatorname{tr}(A \pm B) = \operatorname{tr} A \pm \operatorname{tr} B$$

Scalar multiple: $|kA| = k^n|A|$

Product order: $(m \times n)(n \times p) = m \times p$

Entry: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$

Product determinant: $|AB| = |A||B|$

Trace of product:

$$\operatorname{tr}(AB) = \operatorname{tr}(BA)$$

Identity: $AI = IA = A$

3. Commuting and powers

Commute: $AB = BA$

Anti-commute: $AB = -BA$

Square (if $AB = BA$): $(A \pm B)^2 = A^2 \pm 2AB + B^2$

Difference of squares: $A^2 - B^2 = (A - B)(A + B) \iff AB = BA$

Powers: $A^m A^n = A^{m+n}$, $(A^m)^n = A^{mn}$

Rank one:

$$A = uv^T \Rightarrow A^n = (v^T u)^{n-1} A$$

4. Characteristic equation

Equation: $|A - \lambda I| = 0$

Order 2: $A^2 - (\operatorname{tr} A)A + |A|I = O$

Order 3:

$$A^3 - (\operatorname{tr} A)A^2 + CA - |A|I = O$$

Roots: $\sum \lambda = \operatorname{tr} A$, $\prod \lambda = |A|$

Inverse: $A^{-1} = \frac{1}{q}(pI - A)$ from $A^2 - pA + qI = O$

Division:

$$f(x) = c(x)Q(x) + R(x) \Rightarrow f(A) = R(A)$$

5. Special matrices

Idempotent: $A^2 = A$, $A^n = A$

Idempotent: $(I + A)^n = I + (2^n - 1)A$

Involutory: $A^2 = I$, $A^{-1} = A$, $|A| = \pm 1$

Involutory: $(I + A)^n = 2^{n-1}(I + A)$

Periodic: $A^{k+1} = A$

Nilpotent: $A^m = O$, $(I + A)^n = I + nA + \binom{n}{2}A^2 + \cdots$

6. Transpose, symmetric, skew

Product: $(AB)^T = B^T A^T$

Sum: $(A \pm B)^T = A^T \pm B^T$, $(kA)^T = kA^T$

Symmetric: $A^T = A$; skew: $A^T = -A$

Decomposition: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

Skew, odd order: $|A| = 0$

Quadratic form: $X^T AX = \sum a_{ij}x_i x_j$

7. Orthogonal

Definition: $AA^T = A^T A = I$

Inverse: $A^{-1} = A^T$, $|A| = \pm 1$

Trace:

$$\operatorname{tr}(AA^T) = \sum a_{ij}^2$$

Rotation: $R(\alpha)R(\beta) = R(\alpha + \beta)$, $R(\theta)^n = R(n\theta)$

Permutation: $P^T P = I$, $P^k = I$ (k = lcm of cycles)

8. Adjoint

Definition: $\operatorname{adj} A = [C_{ij}]^T$

Basic:

$$A \operatorname{adj} A = \operatorname{adj} A\, A = |A|I$$

Order 2:

$$\operatorname{adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Determinant: $|\operatorname{adj} A| = |A|^{n-1}$

Double adjoint:

$$\operatorname{adj}(\operatorname{adj} A) = |A|^{n-2} A$$

Double adjoint:

$$|\operatorname{adj}(\operatorname{adj} A)| = |A|^{(n-1)^2}$$

Product:

$$\operatorname{adj}(AB) = \operatorname{adj} B \operatorname{adj} A$$

Power:

$$\operatorname{adj}(A^m) = (\operatorname{adj} A)^m$$

Scalar:

$$\operatorname{adj}(kA) = k^{n-1} \operatorname{adj} A$$

Transpose:

$$\operatorname{adj} A^T = (\operatorname{adj} A)^T$$

9. Inverse

Formula: $A^{-1} = \frac{\operatorname{adj} A}{|A|}$, $|A| \neq 0$

Order 2:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Determinant: $|A^{-1}| = \frac{1}{|A|}$

Double inverse: $(A^{-1})^{-1} = A$

Product: $(AB)^{-1} = B^{-1}A^{-1}$

Transpose: $(A^T)^{-1} = (A^{-1})^T$

Power: $A^{-n} = (A^{-1})^n$

Scalar: $(kA)^{-1} = \frac{1}{k}A^{-1}$

Cancellation: $AB = AC \Rightarrow B = C$ if $|A| \neq 0$

Zero product: $AB = O$, $|A| \neq 0 \Rightarrow B = O$

10. Operations, systems, rank

Rows: $[A \,|\, I] \to [I \,|\, A^{-1}]$

Columns:

$$\begin{bmatrix} A \\ I \end{bmatrix} \to \begin{bmatrix} I \\ A^{-1} \end{bmatrix}$$

System: $AX = B$, $X = A^{-1}B = \frac{(\operatorname{adj} A)B}{|A|}$

Cramer: $x_i = \Delta_i / \Delta$

Inconsistent: $|A| = 0$, $(\operatorname{adj} A)B \neq O$

Homogeneous: $AX = O$ has non-trivial solution iff $|A| = 0$

Matrix equations: $AX = B \Rightarrow X = A^{-1}B$; $XA = B \Rightarrow X = BA^{-1}$

Rank: order of the largest non-singular sub-matrix